Calibration Windows of Toroidal HRR Substrates: Scaling Laws and Empirical Limits for LLM-Compiled Group Reasoning

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Abstract

We characterise toroidal holographic reduced representation (HRR) substrates as verifiers for LLM-compiled symbolic computation, focusing on their scaling behaviour for finite-group composition. We give two structural results: (i) a two-codebook HRR construction resolving the self-loop catastrophe of single-codebook DFA embeddings, with capacity bound $d \geq C\, mk \log(m/\delta)$ and empirical constant $C \in [1, 2]$ validated to $m = 3000$; (ii) a regular-representation embedding of non-abelian D_n via Peter-Weyl, with substrate-level non-commutativity and 100% one-step retrieval at $d = 1024$ for $n \in \{6, 8, 12\}$ up to depth $K = 100$. We then identify three operational regimes of composition separated by $K/\sqrt{D} \approx 1/2$, derive the calibrated-regime margin formula $m(K) \approx 1 - C(V)\sqrt{KV/D}$ (falsifying the Gumbel-asymptotic prediction for $C(V)$), and obtain the design rule $D^* \geq \big(C(V)/(1-m^*)\big)^2 KV$ validated at $V = 16$ against a substrate-only dimension sweep. A crossover formula $L^*(K, V, D)$ frames when the LLM compile step, versus the substrate, is the binding constraint. Empirically, at 0.5–1.5B parameters: compile learning of a multi-atom DSL reaches 82–93% semantic correctness; best-of-32 sampling with the substrate as inference-time verifier matches SFT-greedy on the hardest cross-group split without training; and a controlled RLVR-with-substrate-reward versus SFT comparison at matched compute yields a null result, interpreted via L^* as a below-crossover regime. The substrate-augmented LLM paradigm is operationally bounded but well-defined; we delineate its operating envelope and the axes along which its advantage is expected to grow.

1 Introduction

Ask a frontier language model for the parity of a 50-bit string, or for $g_1 g_2 \cdots g_{50}$ in the dihedral group D_8, and it will fail reproducibly—not stochastically, but at a problem-dependent horizon beyond which accuracy collapses. The same model handles analogy, narrative composition, and chain-of-thought arithmetic of comparable surface complexity. The discriminator is architectural: tasks admitting a closed-form algebraic reduction fit into a transformer's attention circuit, while tasks requiring deep iteration of a rule with substrate-local state exceed the model's arithmetic precision and positional capacity [5, 9, 13, 20].

One response to this gap is to augment the language model with an algebraic substrate that executes composition exactly. Holographic reduced representations [19], and more broadly hyperdimensional computing [11], provide a family of such substrates: a random unit-modulus phase codebook on the complex torus $U(1)^D$ with elementwise Hadamard binding and nearest-neighbour cleanup. Such substrates admit natural group-equivariant extensions [1, 3] and have been used as symbolic memories in cognitive architectures [6]. What has been largely absent from the literature

is a *scaling theory* for the regime where the substrate is coupled to an LLM: how does the substrate dimension D need to scale with the composition depth K, the group cardinality V, and the target fidelity m^*? When does the LLM compile-step become the binding constraint, and when does the substrate?

This paper answers these questions in the setting of finite-group composition, where we can derive and empirically validate an explicit scaling law and its operational design rule. Our contributions are:

C1. A two-codebook HRR fix for the self-loop catastrophe of finite-state-automaton embedding, with capacity bound $d \geq C\, mk \log(m/\delta)$ (Theorem 1) and empirical constant tighter than the Hoeffding analysis predicts.

C2. A non-abelian regular-representation HRR embedding for the dihedral groups D_n (Theorem 2), with strict $T_s T_r \neq T_r T_s$ by construction and stable retrieval through $K = 100$ at $d = 1024$.

C3. A three-regime operational map of HRR substrate composition with breakpoint at $K/\sqrt{D} \approx 1/2$, and the calibrated-regime margin formula $m(K) \approx 1 - C(V)\sqrt{KV/D}$ (Theorem 3), falsifying the asymptotic Gumbel prediction on V-dependence.

C4. An operational design principle $D^* \geq \big(C(V)/(1-m^*)\big)^2\, K\, V$ (Corollary 4) validated at $V = 16$ against a substrate-only dimension sweep.

C5. An LLM-substrate crossover formula $L^*(K, V, D)$ (Corollary 5) framing when the LLM compile-step is the bottleneck versus when the substrate becomes limiting; used to interpret the null result of Section 7 as a below-crossover regime.

C6. An empirical study of the hybrid system at 0.5−1.5B parameter scale: compile-learning of a multi-atom DSL reaching 82–93% semantic correctness, best-of-32 sampling with substrate verification matching SFT-greedy on the hardest split without training, and a controlled RLVR-versus-SFT null at matched compute.

The remainder of the paper is organised as follows. Section 2 fixes notation. Section 3 establishes the two-codebook construction (Theorem 1) and Section 4 the non-abelian embedding (Theorem 2). Section 5—the paper's intellectual core—characterises the three regimes, derives the calibrated-regime margin formula (Theorem 3), validates the design rule, and documents the V-dependence deviation from Gumbel asymptotics. Section 6 reports the LLM-coupled empirical study, and Section 7 the RLVR null result. Section 8 situates the work relative to the HDC, geometric deep learning, LLM scaling, and verifier-reinforcement learning literatures. Section 9 and Section 10 discuss implications and limits.

Position relative to adjacent literatures. Classical HDC and VSA theory [11, 19, 24] provides capacity bounds for *memory retrieval*; we provide bounds for *iterated composition*, where the substrate is coupled to a compilation process and the relevant depth parameter K is set externally. Geometric deep learning [1, 3] bakes group equivariance into learned model weights via convolution; we encode group action into fixed operators and give explicit dimension rules. LLM scaling laws [8, 12] address the model-parameter axis alone; our L^* formula quantifies a bottleneck transition between the LLM and a separate substrate axis that those analyses do not resolve.

2 Background and notation

Toroidal HRR. Fix an ambient complex dimension $D \in \mathbb{N}$. A codebook of size V is a matrix $\mathrm{CB} \in U(1)^{V \times D}$ whose rows are independent draws from a unit-modulus phase distribution: $\mathrm{CB}[v, j] = e^{i\phi_{v,j}}/\sqrt{D}$ with $\phi_{v,j} \sim \mathrm{Unif}[0, 2\pi)$. Elementwise (Hadamard) binding is denoted $\odot$; it is commutative and satisfies $u \odot \bar{u} = \mathbf{1}$. Cleanup of a noisy retrieval $r \in \mathbb{C}^D$ against the codebook returns $\arg\max_v \mathrm{Re}\langle \mathrm{CB}[v], r\rangle$. The cleanup margin $m(r) := \mathrm{top}_1(r) - \mathrm{top}_2(r)$ serves throughout as a confidence diagnostic.

Group action via regular representation. Given a finite group G with $|G| = V$, we associate an independent random codebook $\mathrm{CB}_G \in U(1)^{V \times D}$ and define, for each $g \in G$, the *transition operator*

$$T_g \;:=\; \sum_{i=0}^{V-1} \mathrm{CB}_G[g \cdot i] \otimes \mathrm{CB}_G[i]^* \;\in\; \mathbb{C}^{D \times D}, \tag{1}$$

where $\otimes$ denotes the outer product and $*$ complex conjugation. Applied to $v = \mathrm{CB}_G[j]$, we recover $T_g\, v \approx \mathrm{CB}_G[g \cdot j]$ up to crosstalk of order $O(\sqrt{V/D})$. For non-abelian G, the operators generally fail to commute: $T_s T_r \neq T_r T_s$ by construction, a fact we exploit in Section 4.

Compile-learning interface. A compiler (typically an LLM) emits a finite DSL token sequence that a deterministic parser maps to a substrate program. The substrate executes the program and returns (i) a decoded answer and (ii) a cleanup margin. The LLM may use the margin as an inference-time confidence signal or as a training reward. We consider three distinct operating modes: direct inference (greedy decoding plus substrate execution), best-of-N sampling with substrate verification (Section 6.2), and reinforcement learning with substrate-derived reward (Section 7).

Notation summary. D is the substrate (embedding) dimension; K the composition depth; V the codebook size, typically equal to group cardinality $|G|$; L the LLM parameter count; m^* the target cleanup margin; $C(V)$ the empirically calibrated crosstalk constant; $\varepsilon_{\mathrm{mach}}$ the floating-point machine epsilon; $\tilde{\varepsilon}$ the per-step approximation residual of a trained neural primitive; $\varepsilon_{\min} = 2\pi/V$ the decoder half-gap.

3 Two-codebook HRR for finite-state automata

3.1 The single-codebook self-loop catastrophe

The natural HRR embedding of a deterministic finite-state automaton (Q, Σ, δ) uses a single state codebook S and an alphabet codebook A:

$$M \;=\; \sum_{q \in Q} \sum_{\sigma \in \Sigma} S[\delta(q, \sigma)] \odot \overline{S[q]} \odot \overline{A[\sigma]}. \tag{2}$$

Retrieval of a transition proceeds by binding M with $S[q] \odot A[\sigma]$ and running cleanup against S. This construction is catastrophically broken whenever the automaton contains a self-loop, $\delta(q, \sigma) = q$: the corresponding term in (2) reduces to $S[q] \odot \overline{S[q]} \odot \overline{A[\sigma]} = \mathbf{1} \odot \overline{A[\sigma]} = \overline{A[\sigma]}$, contributing a *deterministic* signal whose magnitude equals the diagonal retrieval signal. Cleanup can no longer distinguish the true target from any candidate whose self-loop on σ dominates the memory. Empirically, at $m = 100$, $|\Sigma| = 2$, and d up to 24000 the single-codebook accuracy plateaus near 30%, with the failures traceable to three self-loops each contributing magnitude 1.00 to incorrect candidates.

3.2 Two-codebook construction

The repair is to use two independent state codebooks S_{tgt} and S_{src}:

$$M \;=\; \sum_{q\in Q}\sum_{\sigma\in\Sigma} S_{\text{tgt}}[\delta(q,\sigma)] \odot \overline{S_{\text{src}}[q]} \odot \overline{A[\sigma]}. \tag{3}$$

The self-loop term $S_{\text{tgt}}[q] \odot \overline{S_{\text{src}}[q]}$ now equals a product of two independent random-phase vectors, distributed identically to a fresh random phase vector. Every cross-term is noise; only the diagonal contributes signal to the correct target.

Theorem 1 (Two-codebook HRR capacity)**.** *Let $\mathcal{A} = (Q, \Sigma, \delta)$ be a DFA with $|Q| = m$ and $|\Sigma| = k$. Let $\delta \in (0, 1/2)$. Construct the memory M as in (3) with independent random unit-phase codebooks $S_{\text{tgt}}, S_{\text{src}} \in U(1)^{m\times d}$ and $A \in U(1)^{k\times d}$. Then for any query (q, σ), the probability that cleanup of $r := M \odot S_{\text{src}}[q] \odot A[\sigma]$ against S_{tgt} returns $\delta(q, \sigma)$ is at least $1 - \delta$, provided*

$$d \;\geq\; C \cdot mk \,\log(m/\delta), \qquad C \leq 16. \tag{4}$$

Proof sketch. Expanding the query, the correct-target score is $1+\eta^*$ where η^* is a sum of $(mk-1)\cdot d$ independent unit-modulus terms with mean zero and variance bounded by $1/2$ per coordinate. Each incorrect-target score is a sum of $mk\cdot d$ such terms, also mean-zero with variance $1/2$ per coordinate. A sufficient condition for cleanup to succeed is that $|\eta| < 1/2$ for every candidate. Applying Hoeffding's inequality to each candidate and a union bound over m candidates gives $\Pr[\text{failure}] \leq 2m\exp(-d\,/\,8mk)$. Setting this $\leq \delta$ yields $d \geq 8mk\ln(2m/\delta)$, and bounding $\ln(2m/\delta) \leq 2\log(m/\delta)$ for $\delta \leq m$ gives $d \geq 16\,mk\log(m/\delta)$, the stated $C = 16$. This Hoeffding bound is loose: the actual unit-modulus tail is sub-Gaussian with strictly smaller variance proxy, and empirically (Section 3, Table 1) the effective constant is $C\in[1,2]$. A full calculation is given in Appendix A. □

3.3 Empirical scaling

Table 1 reports the minimum dimension $d_{\min}$ achieving $\geq 95\%$ retrieval accuracy on 30 random queries per configuration, with $|\Sigma|=2$, $\delta=0.05$, and m ranging over two orders of magnitude. The empirical ratio $d_{\min}/d_{\text{pred}}$ settles near 0.5 from $m = 100$ onward (d_{pred} uses $C = 4$ as a reference target, an order of magnitude tighter than the proven $C\leq 16$), indicating that the effective constant is $C\approx 1-2$ in the operating regime of interest. The slack reflects the tighter tail behaviour of unit-modulus random variables: the complex-phase case is closer to Bernoulli than Gaussian, for which the union bound is looser. At the boundary of the explored range ($m = 3000$, $d = 60{,}000$) only 95% retrieval is reached at $d_{\min}/d_{\text{pred}} = 0.23$; 100% requires $d = 140{,}000$, consistent with the $C \approx 1$ asymptote.

3.4 Depth robustness

At $d = 4\,d_{\min}$, retrieval accuracy remains $\geq 95\%$ at composition depth $K \in \{1, 10, 100\}$ for $m \in \{100, 1000\}$, consistent with the expected per-step error $\ll 1/K$ guaranteed by the capacity margin. Operationally, setting $d = O(mk\log(mK/\delta))$ suffices for deep walks; the logarithmic penalty in K is mild. This behaviour carries over to the non-abelian case analysed next.

m	d_{pred}	$d_{\min}$	$d_{\min}/d_{\text{pred}}$	accuracy at $d_{\min}$
10	424	128	0.30	96.7%
30	1 536	384	0.25	96.7%
100	6 081	3 040	0.50	100%
300	20 879	10 439	0.50	100%
1000	79 228	39 614	0.50	100%
3000	264 051	60 000	0.23	95.0% (100% at $d = 140\,000$)

Table 1: Two-codebook HRR scale sweep. d_{pred} uses the theoretical constant $C = 4$; the empirical column reports the smallest d reaching 95% retrieval.

4 Non-abelian group action via Peter-Weyl

4.1 Dihedral groups and their regular representation

The dihedral group D_n of order $2n$ consists of rotations r^k and reflections sr^k for $k \in \{0, \ldots, n-1\}$, with relations $r^n = e$, $s^2 = e$, $sr = r^{-1}s$. We index elements as $i = 2k + t$ with $t \in \{0, 1\}$ identifying rotation vs. reflection. Right-multiplication rules are

$$r^k \cdot r = r^{k+1 \bmod n}, \qquad r^k \cdot r^{-1} = r^{k-1 \bmod n},$$
$$r^k \cdot s = sr^{(-k) \bmod n}, \qquad sr^k \cdot s = r^{(-k) \bmod n}.$$

The generator set $\{r, r^{-1}, s\}$ spans the full group.

4.2 HRR lift and non-commutativity

For a random unit-phase codebook $\text{CB} \in U(1)^{2n \times D}$, the transition operator T_g of (1) is a sum of $2n$ rank-one outer products. Applied to $v = \text{CB}[j]$, it recovers $\text{CB}[g \cdot j]$ up to crosstalk. Non-commutativity is structural: for any $n \geq 3$, direct computation shows $T_s T_r \cdot \text{CB}[0]$ recovers sr^{n-1} while $T_r T_s \cdot \text{CB}[0]$ recovers sr^1. Since $\text{CB}[\text{index of } sr^{n-1}] \neq \text{CB}[\text{index of } sr^1]$ as random phase vectors, the operator product is non-commutative in the substrate, not merely in the underlying group.

Theorem 2 (Non-abelian regular-representation HRR)**.** *For the dihedral group D_n with $|D_n| = 2n$ and failure tolerance δ, the HRR transition operators of (1) with $d \geq C\,(2n)\log(2n/\delta)$, $C \leq 16$, execute any K-fold composition $g_1 \cdots g_K$ with per-step crosstalk $O(\sqrt{|G|/d})$ and overall retrieval accuracy $\geq 1 - \delta - O(\sqrt{K|G|/d})$ in the calibrated regime $K/\sqrt{d} < 1/2$ (Section 5.3). The substrate operators T_g inherit the non-commutativity of G by construction.*

Proof sketch. The one-step bound follows from Theorem 1 applied to the DFA $(D_n, \{r, r^{-1}, s\}, \cdot)$; the per-step crosstalk is the same unit-modulus tail. The K-fold bound follows by iteration: each application adds a term of order $|G|/d$ to the effective crosstalk variance, which compounds linearly in K in the calibrated regime (Theorem 3); the corresponding tail bound is therefore $\sqrt{K|G|/d}$ rather than the $K\sqrt{|G|/d}$ that a naive union bound over steps would yield. Non-commutativity follows from the random-phase construction: distinct group elements receive independent unit-phase codewords by construction, and coincide with probability zero. See Appendix A for details. □

D	AUROC(margin, correct)	mean margin in-capacity
1024	0.867	7×10^{-7}
2048	0.951	7×10^{-7}
4096	0.951	7×10^{-7}

Table 2: Margin as a correctness signal on a mix of in-capacity and out-of-capacity DFA embeddings. At $D \geq 2048$ the discrimination is near-saturated; the in-capacity regime retains a margin that is essentially noise-level of the random-phase ensemble.

4.3 Empirical validation

For $n \in \{6, 8, 12\}$ at $d = 1024$, we run 50 random walks per $K \in \{1, 2, 5, 10, 20, 50, 100\}$. Across all configurations, one-step retrieval accuracy is 100%; mean cleanup margins range from 0.81 (at $K = 100$, worst case) to 0.96 (at $K = 1$). The margin decay with K follows the formula derived in Section 5.4; no regime transition occurs at these operating points.

4.4 Connection to geometric deep learning

The operator family $\{T_g\}_{g \in G}$ constitutes a regular representation of G on $\mathbb{C}^D$: it satisfies $T_{gh} \approx T_g T_h$ up to crosstalk of order $\sqrt{|G|/d}$. Geometric deep learning [1, 3] bakes group equivariance into learned model weights via group convolution; our approach instead encodes the group action into fixed substrate operators computed from a random codebook. The two approaches are complementary: convolutional equivariance constrains the learned model, substrate operators provide a fixed external reference frame that the model addresses through a compile interface.

5 Scaling laws and operational design principles

This section contains the paper's main scaling-theoretic results. We work throughout on the iterated group-composition task of Section 4 and study how the cleanup margin depends on the substrate dimension D, the composition depth K, and the codebook size $V = |G|$.

5.1 Margin as a confidence signal

We first verify that the cleanup margin discriminates correct from incorrect retrievals. We mix in-capacity instances ($m \in \{5, 10, 20, 50\}$, $k = 2$, $V \in \{5, 10, 20, 50\}$) with out-of-capacity instances ($m \in \{300, 500, 1000\}$, $k = 4$), and compute the area under the ROC curve of margin as a predictor of correctness. Table 2 reports AUROC values across three substrate dimensions.

The discrimination is already near-saturated at $D = 2048$. Operationally, this means margin-gated inference—rejecting queries with margin below a threshold—is a calibrated uncertainty mechanism without any additional training: a free metacognitive primitive.

5.2 Margin as an Occam regulariser

Within the calibrated regime—defined precisely in Section 5.3—the cleanup margin also encodes the length of the composition that produced the current state. We sweep walk length $K \in [1, 12]$ on D_n for $n \in \{4, \ldots, 8\}$ and $D = 512$, and compute Pearson correlation across (n, K) cells. The aggregated correlation is $\rho(K, m) = -0.61$, with cell-level values ranging from -0.99 to $+0.71$ and coefficient of variation 6.6% across walks with the same target. (K here denotes the walk length,

i.e., the composition depth of Section 5.3; the symbol L is reserved throughout for LLM parameter count.)

This observation has a clean information-theoretic interpretation. Under the calibrated regime, the margin decrement after K compositions is $\sim C(V)\sqrt{KV/D}$ (Theorem 3). The walk length K coincides with the description length of the composition in the generator alphabet, and shorter descriptions admit tighter cleanup. This recovers the Solomonoff/Rissanen minimum description length principle [10, 22, 25] as a physical property of the substrate, rather than an externally imposed regulariser. We note that the signal is weak (CV = 6.6%), making it useful as a secondary tie-breaker in search but not as a primary loss. We explicitly avoid framing this observation in the language of Friston's free-energy principle; that framework is too general to be informative for the specific claim, which MDL captures directly.

5.3 Three regimes of substrate composition

The margin decay formula derived in Section 5.4 holds only within a bounded region of the (K, D, V) space. Empirically, we observe three operational regimes, separated by breakpoints in $K/\sqrt{D}$:

Calibrated ($K/\sqrt{D} < 1/2$).
The crosstalk dominates, $\|v\|^2$ remains $O(1)$, and the margin follows the formula of Theorem 3. Accuracy is 100% up to numerical precision; the substrate behaves as an exact group-action executor.

Borderline ($1/2 \leq K/\sqrt{D} \leq 1$).
The non-exact unitarity of T_g and crosstalk accumulation start to compete. Margin variance grows rapidly; the signal becomes unreliable for fine-grained ranking.

Broken ($K/\sqrt{D} > 1$).
The substrate state diverges: $\|v\|^2$ grows compound, accuracy collapses towards chance ($1/V$), and the margin loses informational content entirely.

We illustrate the broken regime with a direct test at $n = 8$, $D = 512$, $K = 100$ ($K/\sqrt{D} \approx 4.4$): mean margin climbs to 332.7 (norm explosion) and accuracy collapses to 28%, vs. the theoretical 100% inside the calibrated regime.

Remark. The three-regime structure is, to our knowledge, not present in the HDC literature, which works in the $K \leq 10$ regime where the broken transition is invisible. It is also absent from geometric deep learning, where exact equivariance is imposed by construction and the calibrated/broken distinction does not arise. LLM scaling laws [8, 12] do not parametrise a substrate dimension and thus cannot express the transition.

5.4 Calibrated-regime margin formula

Within the calibrated regime, the crosstalk analysis yields a closed form for the expected cleanup margin. Consider a single walk of length K through random-phase operators $T_{g_1}, \ldots, T_{g_K}$ starting from $v_0 = \mathrm{CB}[0]$.

Derivation. The correct-target score approximates 1 (the diagonal signal is conserved up to numerical drift in the calibrated regime). The off-target scores, for $j \neq$ target, behave as sums of K independent unit-modulus crosstalk contributions, each with $\mathrm{Var}(\mathrm{Re}) = 1/(2D)$ per coordinate. Aggregating over K steps and D coordinates yields variance $K/(2D)$ per off-target. The cleanup

D	$K/\sqrt{D}$	m_{obs}	m_{pred}	$P(m > 0.9)$	$\|v\|^2$
256	0.750	0.698	0.749	0.150	4.76
384	0.612	0.812	0.795	0.250	3.70
512	0.530	0.822	0.822	0.230	2.41
768	0.433	0.892	0.855	0.395	2.42
1024	0.375	0.907	0.874	0.520	1.55
1614	**0.299**	**0.944**	**0.900**	**0.775**	**1.35**
2048	0.265	0.931	0.911	0.795	1.41
3072	0.217	0.904	0.927	0.540	1.22
4096	0.188	0.933	0.937	0.965	1.10

Table 3: Validation of the D^* formula at $V = 16$, $K = 12$. The highlighted row is the predicted $D^* = 1614$ at target margin $m^* = 0.9$. Accuracy is 100% at every D; the discriminating quantity is the margin and the distribution of confident retrievals $P(m > m^*)$.

margin equals the gap to the maximum of $V - 1$ such off-target scores. Modelling the max by Gumbel asymptotics for independent real Gaussians with variance $\sigma^2 = K/(2D)$ gives

$$\mathbb{E}[m(K, V, D)] \;\approx\; 1 \;-\; \sqrt{\tfrac{\log(V-1)\,K}{D}} \;=\; 1 \;-\; C_{\text{Gumbel}}(V)\,\sqrt{KV/D}, \tag{5}$$

with $C_{\text{Gumbel}}(V) = \sqrt{\log(V-1)/V}$. For $V = 16$, $C_{\text{Gumbel}} \approx 0.41$ asymptotically, or ≈ 0.37 with the standard Gumbel mean correction $-\gamma/\sqrt{2\log(V-1)}$.

Theorem 3 (Calibrated-regime margin)**.** *For a random unit-phase codebook of size V and composition depth K with $K/\sqrt{D} < 1/2$, the expected cleanup margin after applying transition operators $T_{g_1}, \ldots, T_{g_K}$ to $v_0 \in \mathrm{CB}$ satisfies*

$$\mathbb{E}[m(K, V, D)] \;\approx\; 1 \;-\; C(V)\,\sqrt{K\,V/D}, \tag{6}$$

where $C(V)$ is a V-dependent constant admitting the asymptotic form $\sqrt{\log(V-1)/V}$ but requiring empirical calibration per group cardinality (Section 5.6).

Empirical verification at V=16. Sweeping $D \in \{512, 1024\}$ and $K \in [1, 25]$ within the calibrated window, we find best-fit $C_{\text{emp}} \approx 0.29$ at both D values, with $R^2 \geq 0.97$. The gap to the Gumbel prediction ($\sim 25\%$) is accounted for by (i) target-value growth $\|v\|^2 \neq 1$ caused by non-exact unitarity and (ii) finite-V corrections in the max-of-Gaussians approximation. The form of the scaling is robust; the constant requires calibration.

5.5 Operational design principle: the D^* formula

Inverting (6) yields the substrate-dimension provisioning rule.

Corollary 4 (Calibration-window design rule)**.** *For a task with composition depth K, codebook size V, and target margin m^*, the substrate dimension D should satisfy*

$$D^* \;\geq\; \frac{C(V)^2}{(1 - m^*)^2}\;K\,V. \tag{7}$$

[Figure placeholder: $C(V)$ empirical vs. Gumbel asymptotic, log-log axes, generated from test3i raw data in the reproducibility archive.]

Figure 1: V-dependence of the crosstalk constant $C(V)$. Empirical fit $C(V) \approx 11.2\,V^{-1.23}$ versus Gumbel asymptotic $\sqrt{\log(V-1)/V}$. The functional form $\sqrt{KV/D}$ is preserved; the constant is not predictive from first principles in this regime.

Empirical validation. Table 3 reports a substrate-only sweep at $V=16$, $K=12$, with D ranging from 256 to 4096 and 200 random walks per row. At the predicted $D^*=1614$, the observed mean margin is 0.944 (exceeding the 0.9 target) and the fraction of walks with margin above 0.9 is 77.5%. The formula is conservative by roughly 30%: the empirical D^* at which 90% of walks exceed the target margin is 1000–1290, depending on the fitting rule. Fitting $m(D) = 1 - C_{\text{fit}}\sqrt{KV/D}$ on the safe-window subset gives $C_{\text{fit}} \in [0.23, 0.34]$, within 25% of the Gumbel prediction. Crucially, the formula fails outside the calibration window: at $D=256$ ($K/\sqrt{D}=0.75$, borderline) it under-predicts the observed margin by 0.05, consistent with the regime map of Section 5.3.

5.6 V-dependence and the Gumbel rejection

The asymptotic prediction $C(V) = \sqrt{\log(V-1)/V}$ in Theorem 3 assumes that the $V-1$ off-target scores behave as independent Gaussians. We sweep $V \in \{8, 12, 16, 20, 24, 32, 40\}$ at $D=1024$ and $K=10$, with 300 walks per V, and fit $C(V)$ from the observed margin.

The best-fit empirical form is $C_{\text{emp}}(V) \approx 11.2 \cdot V^{-1.23}$, whereas the Gumbel asymptotic predicts a slope of approximately -0.31 in this range. The empirical slope is four times steeper: $C_{\text{emp}}(8) = 0.83$ versus Gumbel 0.49, and $C_{\text{emp}}(40) = 0.10$ versus Gumbel 0.30. The functional form $\sqrt{KV/D}$ is preserved; the constant is not first-principles predictive.

The likely source of the deviation is that, on the regular representation of D_n, off-target codewords lie on a group orbit and are *not* statistically independent: they share algebraic structure that correlates their crosstalk contributions. At small V (small n) the orbit is tight and correlations are strong, inflating C; at large V the orbit spreads and correlations attenuate, depressing C below the Gumbel baseline. A rigorous theoretical model of these correlations is open; for the purposes of this paper, we treat Theorem 3 as a semi-empirical relation: the functional form is derivation-backed, the constant requires per-V empirical calibration.

6 LLM-substrate hybrid system

We now turn to the system-level study, in which a small pretrained LLM compiles natural-language task descriptions into a substrate DSL. We report three distinct empirical results: compile-learning in few-shot, substrate-verified best-of-N sampling, and SFT scaling across the 0.5B–1.5B parameter range.

Split	N=1	N=8	N=32
ID	8.5%	25.4%	**44.4**%
OODG-clean	2.4%	15.2%	**36.4**%
D_8	2.9%	16.7%	**33.3**%

Table 4: pass@N with substrate-as-verifier on `Qwen2.5-0.5B-Instruct` (no fine-tuning). Headline: pass@32 on D_8 reaches 33.3%, matching SFT-greedy on the same split (see Section 6.3), with no training. The substrate's value comes from aggressive inference-time search: pass@1 is very low, pass@32 is SFT-competitive.

6.1 Compile-learning

We use Qwen2.5-0.5B-Instruct with 5 in-context examples of a DSL over three atom families: a single-atom `shift` operation, a 2-atom `compose(shift, shift)` composition, and a 3-atom `compose(compose(shift, shift), scale)` composition. Each family is defined over D_n for $n \in \{5, \ldots, 25\}$, with randomly chosen integer coefficients. The LLM emits a `\boxed{...}` DSL program; a deterministic parser extracts the arguments and the substrate executes. Semantic correctness requires the execution to match ground truth on 10 random probe values of the variable n.

Across 50 single-atom tasks, 30 2-atom tasks, and 30 3-atom tasks, semantic correctness rates are 82.0%, 93.3%, and 90.0% respectively, all higher than chance by orders of magnitude and stable across seeds. Compositional rate matches or exceeds single-atom rate, indicating that the failure mode is template pattern-matching rather than symbolic computation: the LLM is addressing the substrate, not simulating it.

6.2 Best-of-N with substrate as verifier

Given the base model and the substrate as verifier, we evaluate pass@N accuracy at $N \in \{1, 8, 32\}$ on three splits: ID (D_4, D_6 with $K \in \{3, 5, 7\}$), OODG-clean (D_5, D_7 with $K \in \{3, 5, 7\}$), and D_8 with the same K.

The headline is that pass@32 on D_8 reaches 33.3%, matching the SFT-greedy accuracy of the 0.5B model on the same split (Table 5), with no fine-tuning. The gap between pass@1 and pass@32 is large—a factor of $\sim 10\times$ on D_8 and the OODG-clean split—indicating that the substrate's contribution here is as a high-quality verifier over a diverse candidate pool, not as a reranker of an already-correct distribution. This is a substitution of inference-time compute for training-time compute, in line with classical verifier-RL analyses [2, 14] but for an algebraic rather than a learned verifier.

6.3 SFT scaling from 0.5B to 1.5B

We fine-tune LoRA adapters ($r = 8$, $q/k/v/o$ projections) on Qwen2.5-0.5B and Qwen2.5-1.5B over the same 200 supervised DSL examples, same learning rate (2×10^{-4}), same 200 training steps. Table 5 reports greedy accuracy across the three splits.

The 40-percentage-point gain on the D_8 cross-group split from scaling alone is strong evidence that, at the 0.5B scale, the LLM compile step rather than the substrate was the binding constraint. This observation motivates the crossover formula of the next subsection.

Split	BASE	SFT-0.5B	SFT-1.5B
ID (train distribution)	17%	89%	**100%**
OODG-clean	18%	55%	**91%**
D_8 (cross-group)	13%	33%	**73%**

Table 5: SFT at 0.5B vs. 1.5B parameters, same dataset, same optimiser. Tripling the parameter count resolves 40pp on the hardest cross-group split and saturates in-distribution accuracy.

6.4 LLM-substrate crossover

Combining the substrate ceiling from Corollary 4 with an empirical estimate of compile accuracy as a function of LLM parameter count, we obtain:

Corollary 5 (LLM-substrate crossover)**.** *Let* $\mathrm{LLM}_{\mathrm{cmp}}(L, V)$ *denote the compile accuracy of a* L*-parameter LLM on a* V*-ary DSL task. The joint hybrid-system accuracy satisfies*

$$\mathrm{acc}(K, V, D, L) \;\approx\; \min\Big(\mathrm{LLM}_{\mathrm{cmp}}(L, V),\; 1 - C(V)\sqrt{KV/D}\Big). \tag{8}$$

The crossover $L^*(K, V, D)$ *is defined by equality of the two terms.*

For the operating point $K = 12$, $V = 16$, $D = 2048$, the substrate ceiling is ~ 0.91. We estimate $L^* \approx 3-7$B by extrapolating the 0.5B–1.5B compile accuracy curve (17% $\to$ 89% on ID, 13% $\to$ 73% on D_8) to the point where it meets the 0.91 ceiling under two alternative functional forms: logarithmic-in-L (yielding $\sim$ 3B) and Chinchilla-style power law with exponent -0.34 (yielding $\sim$7B).[1] At $L = 1.5$B (the largest scale reported here) the hybrid operates strictly below crossover: the LLM compile step is the bottleneck, and the substrate is not the binding constraint. This fact is the key to interpreting the null result of Section 7.

7 Null result: RLVR vs. SFT at matched compute

We now report a controlled negative result. Under the same task family and parameter scale as Section 6.3, we compare (i) supervised fine-tuning against ground-truth DSL labels, cross-entropy loss, and (ii) reinforcement learning with verifier reward (RLVR): the substrate-derived success signal is used as reward, optimised by REINFORCE with a KL constraint to the SFT reference and a best-snapshot selection over the training run. Compute budget is matched: same number of gradient updates, same LoRA configuration, same 200 tasks.

Table 6 reports the comparison at the 0.5B parameter scale (where the two training regimes are directly comparable at matched compute). SFT outperforms RLVR-best by 26−61pp on every evaluation split. Tracing the RLVR training dynamics, we find that the reward signal transmitted through the LLM policy is dominated by the compile-step stochasticity rather than by the substrate-derived success signal: at the 0.5B scale the policy has significant freedom to produce varied (and often-invalid) DSL, and the KL constraint cannot simultaneously encourage valid structure and search the reward landscape.

[1] The extrapolation rests on two datapoints and is therefore a back-of-the-envelope estimate, not a fitted law; we report the bracket [3, 7]B to indicate its sensitivity to the assumed curve. A third datapoint at 7B would directly falsify or confirm the prediction (see Section 10).

Split	BASE	SFT-0.5B	RLVR-best-0.5B
ID	17%	**89%**	28%
OODG-clean	18%	**55%**	27%
D_8	13%	**33%**	7%

Table 6: Controlled RLVR-vs-SFT at matched compute, 0.5B parameters. SFT dominates by $26-61$pp on every split. Interpretation via the L^* crossover: this operating point is deep below-crossover, where the LLM compile step is the bottleneck and substrate-reward provides no additional leverage over direct supervision.

Interpretation via the crossover. The L^* formula of Corollary 5 identifies L^* for this operating point as $3-7$B, placing both the 0.5B RLVR-vs-SFT comparison above and the 1.5B SFT scaling check of Section 6.3 squarely in the below-crossover regime (see Section 6.4 for the derivation of the L^* interval from the empirical SFT scaling data). In this regime, the binding constraint is the LLM's compile accuracy, not the substrate's composition capacity. RLVR with a substrate-derived reward transmits its signal through the same compile channel that bounds SFT, and pays an additional variance cost from the reward-sampling noise; it therefore cannot improve on SFT. We predict that at $L > L^*$—where the LLM has saturated compile accuracy and the substrate ceiling binds—RLVR should recover its expected advantage. This prediction is testable, though outside the scope of this paper.

Remark. We flag explicitly that the null is specific to a matched-compute comparison at below-crossover scale. It does not rule out RLVR providing value at larger scale, nor does it preclude variants (PPO with tighter KL schedules, process reward shaping as in [14], off-policy RL) from changing the quantitative comparison. What it does rule out is the naive reading "substrate-as-verifier RL beats supervised distillation" at this scale and in this setup.

8 Related work

HRR, VSA, hyperdimensional computing. Plate's original HRR construction [19] established phase-vector binding and cleanup as a symbolic substrate. Kanerva's hyperdimensional computing [11] generalised this to a broader family of distributed representations. Gayler's VSA [7] unified these approaches. Eliasmith's SPA [6] built full cognitive architectures on HRR-like representations. Recent surveys [24] compare modern variants (FHRR, MAP, BSC). Applications include robotics [18] and symbolic reasoning. Our two-codebook fix addresses a specific failure mode (self-loops in DFA embedding) that the classical analyses do not isolate. Our calibration-window scaling and the D^* design principle introduce the LLM-coupled composition-depth axis K, which is not present in the classical capacity analyses (focused on static memory retrieval).

Geometric deep learning and group-equivariant neural networks. Cohen and Welling's group equivariant convolutional networks [3] and the broader geometric deep learning programme [1] build model equivariance into learned weights via group convolution. Maron et al. [15] analyse universality of such networks. Our approach is complementary: we encode group action into fixed substrate operators computed from a random codebook, and we provide explicit scaling rules for the substrate dimension D. The two approaches differ in what is learned (weights vs. none for the substrate) and what is fixed (the group action vs. the architectural symmetry).

LLM scaling laws and compositional generalisation. Kaplan et al. [12] and the Chinchilla work of Hoffmann et al. [8] establish scaling laws for LLM loss as a function of parameters and data. These analyses are orthogonal to the substrate-dimension axis we study. Compositional generalisation in LLMs has been analysed extensively [4, 5, 9, 13, 20]; our contribution in this space is the quantitative bottleneck transition of Corollary 5, which connects LLM scaling and substrate capacity in a single formula.

Tool use, verifier RL, and process supervision. Toolformer [23] demonstrated that LLMs can learn to invoke external tools. Cobbe et al. [2] established verifier training for math word problems; Lightman et al. [14] refined this with process-level reward models. DPO [21] offered an alternative to reward-model RL. Our contribution is the algebraic character of the verifier: the substrate is a deterministic external executor rather than a learned neural model, which gives the verifier a well-defined operating regime (D^* calibration window) and a parametrised crossover to pure-LLM approaches (L^*).

Mechanistic interpretability. Mechanistic interpretability [16, 17] extracts circuit-level explanations from trained models. The HRR substrate is interpretable by construction at the operator level: each T_g corresponds to an exact group action, with cleanup margins providing per-step uncertainty. This is complementary to post-hoc circuit extraction.

9 Discussion

Operational implications. The D^* formula provides, for the first time, a closed-form rule for provisioning an HRR substrate to a task of known composition depth K, codebook size V, and target margin m^*. For a practitioner building a substrate-augmented pipeline, this replaces the "choose D large enough" heuristic with an explicit number, calibrated at $V = 16$ and conservatively bounding the required dimension by $\sim 30\%$. The L^* crossover formula complements this: it identifies when additional LLM compute is productive (below L^*) and when additional substrate dimension is the limiting investment (above L^*).

What this work establishes. We establish (i) a structurally clean HRR construction for DFA embedding that avoids the self-loop catastrophe; (ii) a non-abelian extension to D_n inheriting non-commutativity by construction; (iii) a three-regime scaling map and a calibrated-regime margin formula; (iv) an operational design rule and a bottleneck-transition formula; (v) an empirical study of the LLM-coupled hybrid system at $0.5-1.5$B parameters, including a strong positive (BoN@32 matches SFT-greedy without training) and a controlled null (RLVR does not beat SFT at below-crossover scale).

What this work does not establish. We do not provide benchmark comparisons on standard compositional generalisation tasks (SCAN, COGS, CFQ); the claims are operational (scaling laws, design rules, below-crossover null) rather than community-benchmark claims. We do not test the L^* crossover prediction above $L = 1.5$B; the $3-7$B bracket is an extrapolation. We do not establish the $C(V)$ constant's behaviour outside the $V \in [8, 40]$, $D = 1024$, dihedral-group window in which it is empirically calibrated.

The honest framing. The substrate-augmented LLM paradigm, as characterised here, provides (i) inference-time correctness extraction via BoN plus substrate verifier, without training; (ii) an effective fine-tuning target for small-to-medium scale LLMs; (iii) operational design rules for substrate dimension and bottleneck transition; (iv) a complete scaling characterisation within the calibration window. It does not provide strict superiority over supervised fine-tuning at any tested operating point; nor does it demonstrate general compositional generalisation in language models. These framings are deliberate and empirically grounded.

10 Limitations and future work

- **Compositional benchmark grounding.** SCAN, COGS, and CFQ evaluations of the hybrid system would anchor the theoretical results on community-standard datasets. A SCAN-style compositional split on D_n is the most direct next experiment.
- **Scale $L > 1.5$B.** The L^* crossover prediction is testable only at larger parameter counts; a 7B comparison with matched compute budget for RLVR vs. SFT would directly falsify or confirm the $L^* \in [3, 7]$B bracket of Section 6.4.
- **Theoretical model of V-dependence.** The Gumbel-asymptotic prediction for $C(V)$ is falsified quantitatively; a rigorous model of correlated crosstalk on group orbits—accounting for the regular-representation block structure—is open.
- **Differentiable substrate as a transformer layer.** Engineering challenge requiring substrate-aware loss design; the discrete DSL parsing step is the principal obstacle, and straight-through estimation with grammar-constrained decoding is the most direct candidate approach.
- **Sensory grounding.** The substrate operates on symbolic tokens; grounding to pixels or audio requires a phase encoder (contrastive embedding into $U(1)^D$), untested here.
- **Continuous compact Lie groups.** The substrate construction extends to SO(3), SU(2), and SE(3) via the universal covering maps (quaternions for SO(3)). A pilot study shows diffusive $\sqrt{K}\,\varepsilon_{\text{mach}}$ scaling rather than the linear calibrated-regime scaling studied here; full development is subject of a companion paper.

11 Conclusion

We have characterised the operational envelope of HRR substrates as LLM-compiled verifiers for finite-group composition. Two structural theorems (two-codebook DFA embedding, non-abelian Peter-Weyl) support a semi-empirical scaling law ($m(K) \approx 1 - C(V)\sqrt{KV/D}$), an operational design principle (D^*-formula), and an LLM-substrate crossover formula. Empirical study at 0.5–1.5B parameters yields a strong positive (BoN@32 with substrate verifier matches SFT-greedy without training on the hardest cross-group split), a controlled null (RLVR with substrate reward underperforms SFT at below-crossover scale), and a corresponding interpretation via the L^* formula. The substrate-augmented LLM paradigm is operationally bounded but well-defined, and its operating envelope is now quantified.

Reproducibility statement

All experiments run on a single CPU core or a single consumer GPU (T4) with NumPy, PyTorch, and `transformers`, using the public `Qwen2.5-{0.5B,1.5B}-Instruct` checkpoints. Random seeds

are fixed throughout. Hyperparameters and runtime tables are in Appendix B. Code and raw result files are available at https://doi.org/10.5281/zenodo.[pending].

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A Proofs

Proof of Theorem 1

Let $r := M \odot S_{\text{src}}[q] \odot A[\sigma]$ be the query with M defined by (3). Expanding,

$$\begin{aligned} r &= \sum_{q',\sigma'} S_{\text{tgt}}[\delta(q',\sigma')] \odot \overline{S_{\text{src}}[q']} \odot \overline{A[\sigma']} \odot S_{\text{src}}[q] \odot A[\sigma] \\ &= S_{\text{tgt}}[\delta(q,\sigma)] \;+\; \text{crosstalk}, \end{aligned}$$

where the diagonal term contributes signal and the off-diagonal terms, involving mismatched $(q',\sigma') \neq (q,\sigma)$, contribute noise. Each off-diagonal term is a product of unit-modulus phase vectors, at least one of which is independent of all other factors; its coordinate distribution is therefore uniform phase, with mean 0 and variance 1/2 per coordinate for the real part.

For the cleanup scores $s_c := \text{Re}\langle S_{\text{tgt}}[c], r\rangle/d$:

Signal score. For $c = \delta(q, \sigma)$, $s_c = 1 + \eta_c^*$ where η_c^* is a sum of $(mk-1) \cdot d$ independent mean-zero terms of bounded support, with total variance $(mk-1)/(2d)$. By Hoeffding, $\Pr[|\eta_c^*| > t] \leq 2\exp(-2dt^2/(mk-1))$.

Noise scores. For any $c \neq \delta(q, \sigma)$, $s_c = \eta_c$ is a sum of $mk \cdot d$ independent mean-zero bounded terms, yielding $\Pr[|\eta_c| > t] \leq 2\exp(-2dt^2/(mk))$.

A sufficient condition for cleanup to return the correct target is $|\eta_c^*| < 1/2$ and $|\eta_c| < 1/2$ for all $c \neq \delta(q, \sigma)$. Union-bounding over m candidates, the failure probability is bounded by $2m\exp(-d/(8mk))$ (using the looser noise-score bound for all m contributions, since $mk-1 \leq mk$). Setting this $\leq \delta$ gives $d \geq 8mk\ln(2m/\delta)$. Bounding $\ln(2m/\delta) \leq 2\log(m/\delta)$ for $\delta \leq m$ yields the stated $d \geq 16\,mk\log(m/\delta)$ with constant $C \leq 16$. The bound is loose by a Hoeffding-vs-sub-Gaussian factor and a union-bound factor; the empirical constant (Table 1) is $C \in [1, 2]$. □

Proof of Theorem 2

Theorem 1 applied to the DFA $(D_n, \{r, r^{-1}, s\}, \cdot)$ with $m = 2n$, $k = 3$ gives the one-step bound. For K compositions, by linearity of expectation and the independence structure of the random codebook, the per-step crosstalk variance adds: after K applications, the squared-error of the retrieved state relative to the exact target scales as $KV/(2d)$. In the calibrated regime $K/\sqrt{d} < 1/2$, this remains bounded and the tail (Gaussian) bound on the maximum off-target score gives cleanup success probability $\geq 1 - \delta - O(\sqrt{KV/d})$ (rather than the $K\sqrt{V/d}$ of the naive step-wise union bound). Non-commutativity follows from the construction: distinct group elements are assigned independent random unit-phase codewords, which coincide with probability zero in continuous phase space. □

Derivation of Theorem 3

The argument is sketched in Section 5.4; here we record the intermediate quantities. The diagonal (signal) score is $1 + O(\sqrt{K/D})$ by direct computation on the random-phase ensemble. Each off-target score has variance $\sigma^2 = K/(2D)$ per coordinate after normalisation. Modelling the max over $V-1$ off-targets as the max of $V-1$ independent Gaussians with variance σ^2, the expected max is $\sigma\sqrt{2\log(V-1)} - \sigma\gamma/\sqrt{2\log(V-1)}$ (Gumbel with mean correction $\gamma = 0.5772$). Subtracting from the signal gives the margin formula. Deviations from this prediction are attributed to (i) $\|v\|^2 > 1$ in non-exact unitarity and (ii) correlated crosstalk on the group orbit (Section 5.6). □

B Reproducibility details

Two-codebook scale sweep (Section 3). Script: `sos_paradigm_theorem_a_m3000.py`. DFAs sampled uniformly over transition functions; $|\Sigma| = 2$; $\delta = 0.05$; 30 random queries per DFA; 5 DFAs per m value.

D_n retrieval (Section 4). Script: `sos_nonabelian_dihedral_v2_hrr.py`. $D = 1024$, $K \in \{1, 2, 5, 10, 20, 50, 100\}$, 50 walks per K, 3 seeds.

Margin AUROC (Section 5.1). Script: `test3_margin_calibrate.py`. In-capacity mix $m \in \{5, 10, 20, 50\}$, $k = 2$; out-of-capacity mix $m \in \{300, 500, 1000\}$, $k = 4$; 300 queries per (m, k) cell.

Three-regime sweep and D^* validation (Section 5.5). Script: `test3h_substrate_dim_validation.py`. $V = 16$, $K = 12$, $D \in \{256, 384, 512, 768, 1024, 1614, 2048, 3072, 4096\}$, 200 walks per D.

V-dependence (Section 5.6). Script: `test3i_V_dependence_calibration.py`. $D = 1024$, $K = 10$, $V \in \{8, 12, 16, 20, 24, 32, 40\}$, 300 walks per V.

Compile-learning (Section 6.1). Script: `sos_compile_compositional.py`. Qwen2.5-0.5B-Instruct, 5 in-context examples, greedy decoding, `max_new_tokens`=500, 50 single-atom / 30 2-atom / 30 3-atom tasks, 10 probe values each.

BoN with substrate verifier (Section 6.2). Script: `sos_mve_target_E_bestofn.py`. `Qwen2.5-0.5B-Instru` (no fine-tuning), temperature 1.0, top-p 0.95, $N \in \{1, 2, 4, 8, 16, 32\}$, 18/22/15 tasks for ID/OODG-clean/D_8 splits respectively, 32 samples per task.

SFT scaling (Section 6.3). Scripts: `sos_mve_target_F_sft_1p5b_D2048.py` and the 0.5B variant. LoRA $r = 8$, $q/k/v/o$ projections, learning rate 2×10^{-4}, 200 steps, 200 supervised examples.

RLVR (Section 7). Script: `sos_mve_target_G_grpo_1p5b.py`. REINFORCE with KL constraint $\beta = 0.05$, best-snapshot selection every 10 steps, matched total gradient steps to SFT, same LoRA configuration.

Seeds. All experiments use seed 4242 for the main run; Section 7 reports the best across 3 seeds for fairness to the RLVR arm. Full hyperparameters and raw results are available in the reproducibility archive.

C Ablations

Orthonormal substrate. Replacing the random unit-phase codebook with an orthonormal codebook (Gram-Schmidt of i.i.d. complex Gaussians) results in cleanup margin identically equal to 1 across all tested K and accuracy 100%. This removes the Occam signal of Section 5.2: the length-regularising effect is an emergent property of the random-phase ensemble, not a design choice. Random unit-phase is thus the informative regime; orthonormal substrate is degenerate on this axis.

Compile-prompt sensitivity. The compile-learning results of Section 6.1 are stable within ± 5pp under prompt template variations (minor rewording of the in-context example format). Major template changes (removing the grammar specification) degrade semantic correctness by 20−30pp.

BoN temperature. BoN accuracy is stable for sampling temperatures in $[0.7, 1.2]$; $T = 1.0$ was used as a neutral value and is not hand-tuned.